Exercise 2.27. Prove that every odd integer is the difference of two squares. (For example, $11 = 6^2 - 5^2$.)

Solution.

Let n be any odd integer. Since n is odd, we can write

$$n = 2k + 1$$

for some integer k.

Now observe that

$$(k+1)^2 - k^2$$

is the difference of two squares. Expanding,

$$(k+1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1.$$

But $2k + 1 = n$. Therefore,

$$n = (k+1)^2 - k^2.$$

Hence every odd integer can be written as the difference of two squares:

$$\boxed{n = (k+1)^2 - k^2}.$$

For example, if $n = 11$, then $k = 5$, so

$$11 = 6^2 - 5^2.$$

Thus, $\boxed{\text{every odd integer is the difference of two squares.}}$